Dehghani first described the current incarnation of the data mesh concept as a set of four principles. Throughout this book, we focus on their implementation details. Following is a summary of these four data mesh cornerstones.

1.4.1 Domain-oriented decentralized data ownership and architecture

The first principle is that *data and its relevant components should be owned, maintained, and developed by the people closest to it*; this means the people inside the data's domain. This calls for applying the concepts of domains and bounded contexts to the data world. This also means applying decentralization to ownership and architecture.

The idea is that, just as in the previous example, the domain-internal engineers are responsible for developing data interfaces that allow other users (data scientists, self-service BI users, data engineers, or system developers) to use that domain data. The data product engineers are expected to be experts within a single domain, which minimizes communication problems and misinterpretations of data.

Data should have clear ownership, and it should not be on the centralized level of an organization. We should put that responsibility into the hands of the people closest to it. That might be the domain team in the case of a source-oriented dataset; or it might be a data engineering team, an analytics engineering team, or a data science team for new datasets created out of multiple datasets. It could also be the organizational unit using the data if our dataset is very consumer-oriented.

The *domain* is an area or part of our business. It is a way of slicing and decomposing the company. Quite often our organizational structure resembles business domains. At Messflix, we can, for instance, find domains that distribute content, produce content, and develop a market and brand. See figure 1.9.